

Optimal control problem with state constraints for cancer chemotherapy and treatment optimization.

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joint with: A. Belmiloudi¹ & M. Haddou¹

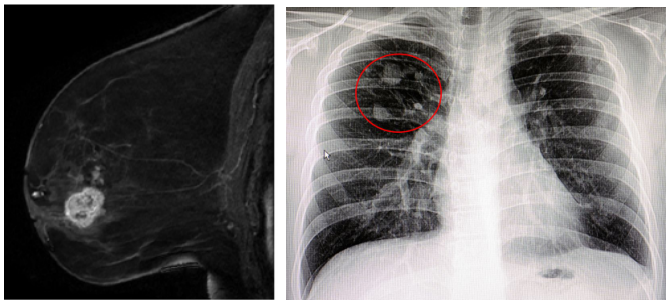
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A tumor is a group of abnormal cells that form a mass:



Source: Ritse M. Mann *et al.*:Breast MRI:State of the art,2019 and www.passeportsante.net

From (CIRC):

- 484000 new cases in France with 191000 deaths in 2022.
- First country in the world in terms of breast cancer index in 2022.

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- Resistance of tumor cells to drugs.
- Tumor resistance after treatments (secondary cancer).

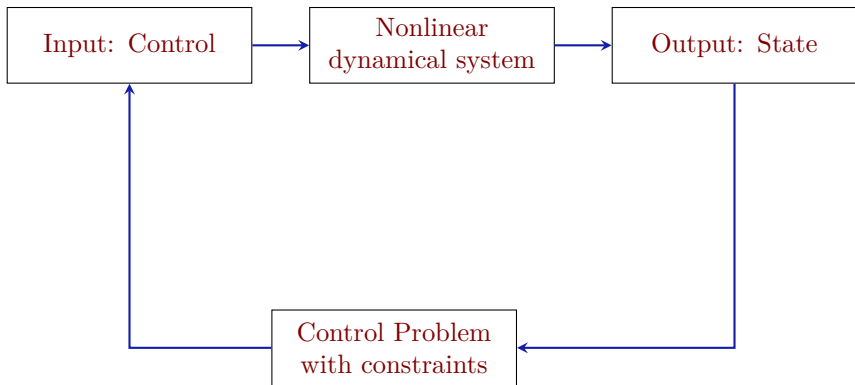
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Mathematical Modeling "+" Optimal control problem with state constraint.

Process



State of the art

Mathematical modeling

- T. Yankeelov, *et al.* 2018; A.M. Jarrett, *et al.* 2019; D.A. Hormuth, *et al.* 2019; Resende, A., *et al.* 2021; Johnson K., *et al.* 2020; A.M. Jarrett, *et al.* 2020; D.A. Hormuth, *et al.* 2021; ...etc

Optimal control

- Segun O., *et al.* 2018; Jarrett A.M., *et al.* 2020; P. Colli, *et al.* 2021; Guillermo L., *et al.* 2021; M. Garavello, *et al.* 2024; ...etc

Goal of the work

Eradication of malignant breast tumors by proposing a new methodology:

- using a formalism of the optimal control problem:

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Eradication of malignant breast tumors by proposing a new methodology:

- using a formalism of the optimal control problem:
- ▶ a constraint on control to limit the acute toxicity of drugs.
- ▶ a constraint on the state to limit the progression of resistant tumors.

Outline of the talk

- 1 Mathematical modeling
- 2 Optimal control problem
- 3 Applications to realistic problems
 - Treatment of breast tumors
 - The influence of state constraint in treatment
 - Treatment of a tumor in a 3D breast domain
 - Treatment of breast tumors in the presence of noise data
 - Treatment of lung tumors
- 4 Conclusion and future work

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$$\left\{ \begin{array}{l} \frac{\partial \mathbf{u}}{\partial t} - \operatorname{div}(\mathbf{D}(t, \mathbf{x}) \nabla \mathbf{u}) + \mathbf{K}_1(t, \mathbf{x}) h_1(\mathbf{u}(t, \mathbf{x})) - \mathbf{K}_2(t, \mathbf{x}) \mathbf{u} + \alpha_0 \phi(t, \mathbf{x}) \mathbf{u} = 0 \\ \quad \text{in } \mathcal{Q} = (0, T) \times \Omega, \\ \quad (\Omega \text{ is the breast or lung domain}) \\ \\ -\mathbf{D}(t, \mathbf{x}) \nabla \mathbf{u} \cdot \mathbf{n} - \mathbf{K}_3(t, \mathbf{x}) h_2(\mathbf{u}(t, \mathbf{x})) = 0 \text{ on } \Sigma = (0, T) \times \partial\Omega, \\ \quad (\mathbf{n} \text{ is the outward normal to } \partial\Omega) \\ \\ \mathbf{u}(t = 0, \mathbf{x}) = u_0(\mathbf{x}) \text{ in } \Omega, \end{array} \right. \quad (1)$$

under the control constraint

$$\mathbf{a}(\mathbf{x}) \leq \phi(t, \mathbf{x}) \leq \mathbf{b}(\mathbf{x}) \quad a.e. (t, \mathbf{x}) \in \mathcal{Q}, \quad (2)$$

with (for $i = 1, 2$)

$$h_i : z \in \mathbb{R} \mapsto h_i(z) = \begin{cases} z^{\rho_i} & \text{if } \rho_i \in \mathbb{N} \setminus \{1\}, \\ z|z|^{\rho_i-1} & \text{if } \rho_i \in \mathbb{R} \setminus \mathbb{N}, \rho_i > 1, \rho_1 + 1 \geq 2\rho_2. \end{cases} \quad (3)$$

Model variables	definitions and dimension
u	Tumor density <i>dimensionless</i>
D	Diffusion function [$mm^2 \cdot d^{-1}$]
K₁	Intrinsic growth rate [d^{-1}]
K₂	Proliferation rate [d^{-1}]
ϕ	Drug concentration [μM]
α_0	Efficacy parameter [$1/d \cdot \mu M$]
K₃	Migration function [$mm \cdot d^{-1}$]
u_0	Initiale tumor density <i>dimensionless</i>
a, b	Positive functions [μM]
T	Total time of treatment in days [d]



ASSUMPTION

(H1) : $u_0 \in W^{r,q}(\Omega)$ with $r = 0, 1$ and $1 < q < \infty$, $\Omega \subset \mathbb{R}^m$ ($m \geq 2$). The coefficients ρ_i for $i = 1, 2$ are fixed such that

$$(I_1) : \begin{cases} \text{(i) if } r = 0, \text{ then } \rho_1 \in]1; \frac{m+2q}{m}] \text{ and } \rho_2 \in]1; \frac{m+q}{m}]. \\ \text{(ii) if } r = 1 \text{ and } q < m, \text{ then } \rho_1 \in]1; \frac{m+q}{m-q}] \text{ and } \rho_2 \in]1; \frac{m}{m-q}]. \\ \text{(iii) if } r = 1 \text{ and } q \geq m, \text{ then } \rho_i \in]1; \infty[\text{ for } i = 1, 2. \end{cases}$$

For the parameter of the model, we assume that: $\mu_1 \geq D(t, x) \geq \mu_0 > 0$, and that $0 < \underline{K}_i \leq K_i(t, x) \leq \overline{K}_i$ for all $(t, x) \in \bar{Q}$, with $\overline{K}_i, \underline{K}_i \in \mathbb{R}_+^*$.



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(H2) : The bounds $\mathbf{a}, \mathbf{b} \in L^{\mathbf{P}^c}(\Omega)$, $0 \leq \mathbf{a} \leq \mathbf{b}$, and $\mathbf{p}_c$ satisfies

$$(I_2) : \begin{cases} \text{(i) for } r = 0 : \\ \mathbf{p}_c \in]\frac{\rho_1}{\rho_1 - 1}; \frac{q\rho_1}{\rho_1 - 1}] \text{ if } \rho_1 \geq q, \text{ and } \mathbf{p}_c \in [\frac{q}{\rho_1 - 1}; \frac{q\rho_1}{\rho_1 - 1}] \text{ if } \rho_1 < q. \\ \text{(ii) for } r = 1 \text{ and } q < m : \\ \mathbf{p}_c \in]\frac{mq}{(m-q)(\rho_1 - 1)}; \frac{q\rho_1}{\rho_1 - 1}] \text{ if } \frac{m}{m-q} < \rho_1, \text{ and } \mathbf{p}_c \in [\frac{\rho_1}{\rho_1 - 1}; \frac{q\rho_1}{\rho_1 - 1}] \text{ if } \rho_1 \leq \frac{m}{m-q}. \\ \text{(iii) for } r = 1 \text{ and } q \geq m, \mathbf{p}_c = \frac{q\rho_1}{\rho_1 - 1}. \end{cases}$$

We denote by $\mathcal{U}_{ad}$ the set of admissible controls defined by

$$\mathcal{U}_{ad} := \left\{ \varphi \in L^{\mathbf{P}^c}(\mathcal{Q}) : \mathbf{a}(x) \leq \varphi(t, x) \leq \mathbf{b}(x), \text{ a.e. } t \in [0; T] \right\}.$$

We suppose that the assumptions H1 and H2 are satisfied.



Theorem

- ① For all $u_0 \in W^{r,q}(\Omega)$ ($r = 0, 1$ and $q > 1$) and $\varphi \in \mathcal{U}_{ad}$, there exists $\underline{\gamma} > 0$, a time $\tau_0 > 0$ and a unique mild solution $\mathbf{u}$ on $[0; \tau_0]$ to (1)-(2)-(3). Moreover, if $\rho_1 + 1 > 2\rho_2$ or $\rho_1 + 1 = 2\rho_2$ with $4\nu_0(q-1) \inf_Q K_1 \geq c_\Omega (\inf_\Sigma K_3)^2 (\rho_2 + q - 1)^2$, then $\mathbf{u} \in C([0, T]; W^{r,q}(\Omega))$ is the classical solution globally defined in all time $t \in (0; T)$ for (1)-(2)-(3).



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- ② Let $\mathbf{u}_1$, resp. $\mathbf{u}_2$ the classical solution with respect to $(u_{0,1}, \varphi_1)$, resp. $(u_{0,2}, \varphi_2)$ belonging to $W^{r,q}(\Omega) \times \mathcal{U}_{ad}$. Then, there exists $C > 0$ such that for all $t \in (0; T)$, $\theta \in [0; \underline{\gamma}]$, $\epsilon \in [0; \theta]$ the following holds

$$\begin{aligned} \|\mathbf{u}_1(t, \cdot) - \mathbf{u}_2(t, \cdot)\|_{W^{2\theta+r,q}(\Omega)} &\leq Ct^{-\theta} \left(\|u_{0,1} - u_{0,2}\|_{W^{r,q}(\Omega)} + \|\varphi_1 - \varphi_2\|_{\mathcal{U}_{ad}} \right), \\ \left\| \frac{\partial \mathbf{u}_1}{\partial t}(t, \cdot) - \frac{\partial \mathbf{u}_2}{\partial t}(t, \cdot) \right\|_{W^{-1,q'}(\Omega)} &\leq Ct^{\epsilon-\theta} \left(\|u_{0,1} - u_{0,2}\|_{W^{r,q}(\Omega)} + \|\varphi_1 - \varphi_2\|_{\mathcal{U}_{ad}} \right). \end{aligned} \quad (4)$$

- ③ For $u_0 \in W^{r,q}(\Omega)$ such that $0 \leq u_0 \leq c_s$ and $\varphi \in \mathcal{U}_{ad}$, the solution $\mathbf{u}$ of (1)-(2)-(3) is bounded in $\mathcal{Q}$ and verifies

$$0 \leq \mathbf{u}(t, \cdot) \leq c_s, \text{ for all } t \in (0; T) \text{ and a.e. in } \Omega, \quad (5)$$

$$\text{where } c_s = \left(\frac{\overline{K_2}}{\underline{K_1}} \right)^{\frac{1}{\rho_1 - 1}}.$$

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In this part, we restrict ourselves to some realistic situations and consider only cases: $\rho_1 \geq 2$, $\rho_2 = 2$, $r = 1$, $q = 2$, $\theta = 0$, $m \leq 3$, and $\mathbf{p}_c \geq 2$ (the existence of such $\mathbf{p}_c$ is garanted when $\rho_1 \geq 2$).

Let $\mathcal{K} := C([0; T])_-$ be the cone of the negative continuous functions on $[0; T]$.

Let define the solution operator

$\mathcal{S} : \phi \in \mathcal{U}_{ad} \mapsto \mathcal{S}(\phi) \in \mathbf{U} := \mathcal{W} \cap C([0, T]; H^1(\Omega))$ such that $\mathcal{S}(\varphi) = \mathbf{u}$ solution of (1)-(2)-(3), where $\mathcal{W} := L^2(0, T; H^1(\Omega)) \cap H^1(0, T; (H^1(\Omega))')$.

The cost functional:

$$\mathcal{J}(\phi) = \frac{1}{2} \int_{\mathcal{Q}} \mathbf{u}^2 d\mathbf{x}dt + \frac{\lambda}{\mathbf{p}_c} \int_{\mathcal{Q}} |\phi|^{\mathbf{p}_c} d\mathbf{x}dt, \quad (6)$$

$$G(\phi) = \|\mathbf{u}(t, \cdot)\|_{L^2(\Omega)}^2 - \xi \leq 0, \quad (7)$$

$$\text{s.t., } \mathbf{u} = \mathcal{S}(\phi).$$

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The control problem:

$$(\mathbf{CP}) \begin{cases} \inf_{\phi \in \mathcal{U}_{ad}} \mathcal{J}(\phi); \\ \text{s.t., } G(\phi) \in \mathcal{K}. \end{cases} \quad (8)$$



Theorem

- ① *The control problem (CP) admits at least one optimal solution $\phi_* \in \mathcal{U}_{ad}$.*
- ② *The solution operator $\mathcal{S}$ is continuously differentiable from $\mathcal{U}_{ad}$ to $\mathbf{U}$ such that the derivative $S'(\phi) : \varphi \mapsto w = S'(\phi) \cdot \varphi$ at point $\phi \in \mathcal{U}_{ad}$ is the unique solution be in $\mathcal{W} \cap C([0; T], L^2(\Omega))$ in the weak formulation sense of the following problem*

$$\begin{cases} \frac{\partial w}{\partial t} - \operatorname{div}(D \nabla w) + (K_1 h'_1(\mathbf{u}) - K_2 + \alpha_0 \phi) w = -\alpha_0 \varphi \mathbf{u} \text{ in } \mathcal{Q}, \\ -D \nabla w \cdot \mathbf{n} - K_3 h'_2(\mathbf{u}) w = 0 \text{ on } \Sigma, \\ w(0, \mathbf{x}) = 0 \text{ in } \Omega. \end{cases} \quad (9)$$



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Moreover, we have the estimates ($\forall \phi \in \mathcal{U}_{ad}$) :

$$\|\mathcal{S}(\phi + \varphi) - \mathcal{S}(\phi) - \mathcal{S}'(\phi) \cdot \varphi\|_{\mathcal{W} \cap C([0, T]; L^2(\Omega))}^2 \leq C \|\varphi\|_{L^{P_c}(\mathcal{Q})}^4, \quad (10)$$

where φ is such that $\phi + \varphi \in \mathcal{U}_{ad}$.

$$\|\mathcal{S}'(\phi_1) \cdot \varphi - \mathcal{S}'(\phi_2) \cdot \varphi\|_{\mathcal{W} \cap C([0, T]; L^2(\Omega))}^2 \leq C \|\phi_1 - \phi_2\|_{L^{P_c}(\mathcal{Q})}^2 \|\varphi\|_{L^{P_c}(\mathcal{Q})}^2 \quad (11)$$

for all $(\phi_1, \phi_2, \varphi) \in \mathcal{U}_{ad} \times \mathcal{U}_{ad} \times L^{P_c}(\mathcal{Q})$.

PART 1.

Theorem

4/4

① The state constraint (7) is qualified if the following condition holds:

$$\left\{ \begin{array}{l} \text{There exists } \phi \in \mathcal{U}_{ad} \text{ such that} \\ \varpi(t) := \mathcal{G}(\phi_*)(t) + \int_{\Omega} \mathbf{u}_*(t, \mathbf{x}) w_{\Phi}(t, \mathbf{x}) \, d\mathbf{x} < 0, \forall t \in (0; T) \\ \mathbf{u}_* = \mathcal{S}(\phi_*) \text{ and } w_{\Phi} \text{ solution of (9) for } \Phi = \phi - \phi_* . \end{array} \right. \quad (12)$$

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- ② Let $\phi_* \in \mathcal{U}_{ad}$ a solution of (CP) and $\mathbf{u}_* = \mathcal{S}(\phi_*)$ such that the condition (12) holds. Then there exist Lagrange multipliers $(\tilde{\mathbf{u}}_*, \mu_*) \in L^2(0, T; H^1(\Omega)) \cap L^\infty(0, T; L^2(\Omega)) \times BV(0, T)_+$, such that $\mu_* \neq 0$, and that $\tilde{\mathbf{u}}_* = \tilde{\mathcal{S}}(\phi_*)$ is the solution of the following adjoint problem

$$\left\{ \begin{array}{l} -\frac{\partial \tilde{\mathbf{u}}_*}{\partial t} - \operatorname{div}(\mathbf{D} \nabla \tilde{\mathbf{u}}_*) + (\mathbf{K}_1 h'_1(\mathbf{u}_*) - \mathbf{K}_2 + \alpha_0 \phi_*) \tilde{\mathbf{u}}_* = \mathbf{u}_* + 2\mathbf{u}_* d\mu_*(t) \text{ in } \mathcal{Q}, \\ -\mathbf{D} \nabla \tilde{\mathbf{u}}_* \cdot \mathbf{n} - \mathbf{K}_3 h'_2(\mathbf{u}_*) \tilde{\mathbf{u}}_* = 0 \text{ on } \Sigma, \\ \tilde{\mathbf{u}}_*(t = T, \cdot) = 0 \text{ in } \Omega. \end{array} \right. \quad (13)$$

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The multiplier μ_* verifies

$$\mathcal{G}(\phi_*) \in \mathcal{K}; \quad \int_0^T \mathcal{G}(\phi_*) d\mu_*(t) = 0; \quad \mu_* \in \mathcal{K}^-; \quad \mu_*(T) = 0, \quad (14)$$

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and we have the following optimality condition

$$\langle \lambda |\phi_*|^{\mathbf{P}c-1} - \alpha_0 \mathbf{u}_* \tilde{\mathbf{u}}_*; \phi - \phi_* \rangle_{L^{\mathbf{P}c}(\mathcal{Q}), L^{\mathbf{P}c}(\mathcal{Q})} \geq 0 \text{ for all } \phi \in \mathcal{U}_{ad} . \quad (15)$$

The penalized cost functional:

$$\mathcal{J}_p(\phi) = \mathcal{J}(\phi) + \frac{\beta_1}{\mathbf{p}_c} \int_{\mathcal{Q}} (\mathbf{a} - \phi)_+^{\mathbf{p}_c} \, d\mathbf{x} \, dt + \frac{\beta_2}{\mathbf{p}_c} \int_{\mathcal{Q}} (\phi - \mathbf{b})_+^{\mathbf{p}_c} \, d\mathbf{x} \, dt + \frac{\beta_3}{4} \int_0^T (\mathcal{G}(\phi))_+^2 \, dt \quad (16)$$

where for all real function u we denote $u_+ = \max(u, 0)$.

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We obtained a control problem without constraint:

$$(\mathbf{PCP}) : \inf_{\phi \in \mathbf{L}^{\mathbf{p}_c}(\mathcal{Q})} \mathcal{J}_p(\phi). \quad (17)$$

PART 1. Necessary optimality conditions (PCP)

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Proposition

- $\mathcal{J}_p$ is continuously differentiable and its gradient is given by

$$\nabla \mathcal{J}_p(\phi) = -\alpha_0 \mathbf{u} \tilde{\mathbf{u}} + \lambda \phi^2 - \beta_1 (\mathbf{a} - \phi_*)^2_+ + \beta_2 (\phi_* - \mathbf{b})^2_+. \quad (18)$$

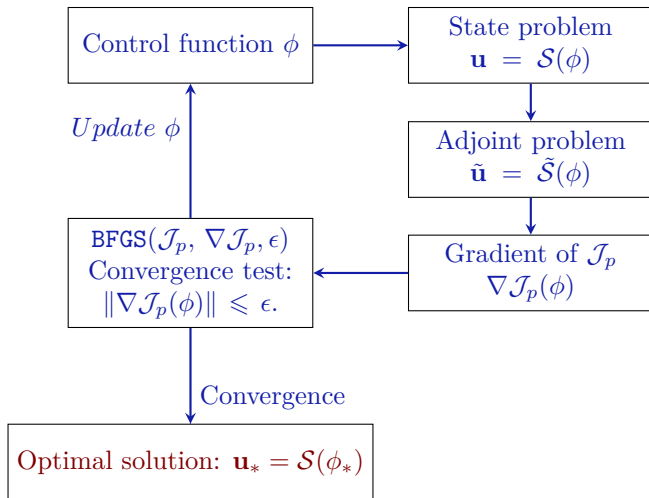
- Let ϕ_* be a solution of (PCP) and $\mathbf{u}_* = \mathcal{S}(\phi_*)$ the associated optimal state. Then $\nabla \mathcal{J}_p(\phi_*) = 0$, (since without constraints) i.e.,

$$\lambda \phi_*^2 - \alpha_0 \mathbf{u}_* \tilde{\mathbf{u}}_* - \beta_1 (\mathbf{a} - \phi_*)^2_+ + \beta_2 (\phi_* - \mathbf{b})^2_+ = 0, \quad (19)$$

where $\tilde{\mathbf{u}}_*$ verifies the following adjoint system

$$\left\{ \begin{array}{l} -\frac{\partial \tilde{\mathbf{u}}_*}{\partial t} - \operatorname{div}(\mathbf{D} \nabla \tilde{\mathbf{u}}_*) + (\mathbf{K}_1 h'_1(\mathbf{u}_*) - \mathbf{K}_2 + \alpha_0 \phi_*) \tilde{\mathbf{u}}_* = \mathbf{u}_* + \beta_3 \mathbf{u}_* \mathbf{G}(\phi_*)_+ \text{ in } \mathcal{Q}, \\ -\mathbf{D} \nabla \tilde{\mathbf{u}}_* \cdot \mathbf{n} - \mathbf{K}_3 h'_2(\mathbf{u}_*) \tilde{\mathbf{u}}_* = 0 \text{ on } \Sigma, \\ \tilde{\mathbf{u}}_*(t = T, \cdot) = 0 \text{ in } \Omega. \end{array} \right. \quad (20)$$

PART 1. Optimization Algorithm



PART 2. Applications to realistic problems

We consider: $\rho_1 = 3$, $\rho_2 = 2$, $\mathbf{p}_c = \frac{2\rho_1}{\rho_1 - 1} = 3$, $r = 1$, $q = 2$, $m = 2$.

Outline of the talk

- 1 Mathematical modeling
- 2 Optimal control problem
- 3 Applications to realistic problems
 - Treatment of breast tumors
 - The influence of state constraint in treatment
 - Treatment of a tumor in a 3D breast domain
 - Treatment of breast tumors in the presence of noise data
 - Treatment of lung tumors
- 4 Conclusion and future work

PART 2-1. Treatment of breast tumors

$$D = \kappa_0 e^{-\lambda_0 \sigma}; \quad \sigma = \sigma_0 e^{-\delta_0 |\mathbf{x}|^2 t}; \quad K_1 = \frac{k_1 - k_2 e^{-\gamma_0 \sigma}}{e^{-\gamma_0 \sigma} (1 - e^{-2\gamma_0 \sigma})}; \quad K_2 = \theta^2 \Psi_1;$$

$$\mathbf{a}(\mathbf{x}) = 0; \quad \mathbf{b}(\mathbf{x}) = \gamma_0 e^{-\gamma_1 [(x-a)^2 + (y-b)^2]}; \quad K_3 = k_3 \Psi_1 / \Gamma; \quad \kappa_0 = d_0 \eta_1;$$

$$\eta_1 = 1 + \varsigma_1 \cos(\varsigma_2 \eta_2); \quad \eta_2 = \varsigma_3 \arctan \left(\frac{y - y_0}{\varsigma_4 + \sqrt{(x - x_0)^2 + (y - y_0)^2}} \right);$$

$$\xi = \|u_0\|_{L^2(\Omega)}^2.$$

Parameters	Values
d_0	$1.5 \text{ mm}^2 \cdot d^{-1}$
σ_0	18 kPa
λ_0	$2 \cdot 10^{-3} \text{ kPa}^{-1}$
k_2	$2.0 d^{-1}$
k_1	$4.0 d^{-1}$
k_3	3.0 mm :
δ_0	$5.55 \cdot 10^{-5} \text{ mm}^{-2} \cdot d^{-1}$
α_0	$1/d \cdot \mu\text{M}$
T	20 days
γ_0	100 μM
γ_1	20 mm^2
θ	2.236 (dimensionless)
ς_1	0.75
ς_2	50
ς_3	2
ς_4	10^{-4}

We used some realistic data from [Lorenzo, G., et al., . 2021.](#)

PART 2-1. Treatment of breast tumors

(1) Case of a single tumor

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2 + (y-y_0)^2]} \text{ in } \Omega, \quad (21)$$

the density of tumor cells centered in (x_0, y_0) .

The discretization and mesh informations:

Parameters	Value	Definitions
Δt	0.2 days (d)	Time step
$\mathbf{x}$	$(x; y)$	Space position
Δx	1 mm	Space discretizations (x)
Δy	1 mm	Space discretizations (y)
n_e	3184	Total mesh elements
n_p	1653	Total mesh points
$(x_0; y_0)$	(10/3; 10/3)	Tumor center
$(a; b)$	(2.9; 10/3)	
ε	0.02	
$\mathcal{P}^1$		Finite elements (Freefem++)

PART 2-1. Treatment of breast tumors

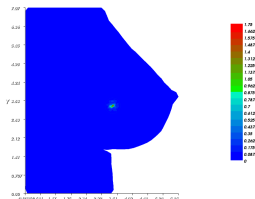
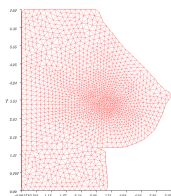
(1) Case of a single tumor

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon} [(x-x_0)^2 + (y-y_0)^2]} \text{ in } \Omega, \quad (21)$$

the density of tumor cells centered in (x_0, y_0) .

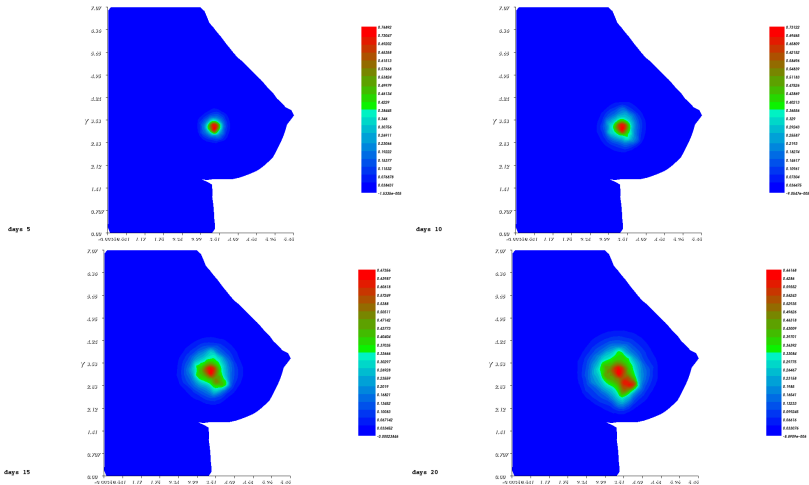
The discretization and mesh informations:

Parameters	Value	Definitions
Δt	0.2 days (d)	Time step
$\mathbf{x}$	$(x; y)$	Space position
Δx	1 mm	Space discretizations (x)
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$(a; b)$	(2.9; 10/3)	
ε	0.02	
$\mathcal{P}^1$		Finite elements (Freefem++)



PART 2-1. Treatment of breast tumors

The density of tumor cells $\mathbf{u}$ without treatment *i.e.*, $\phi = 0$ in (1) at 5^{th} , 10^{th} , 15^{th} , 20^{th} days

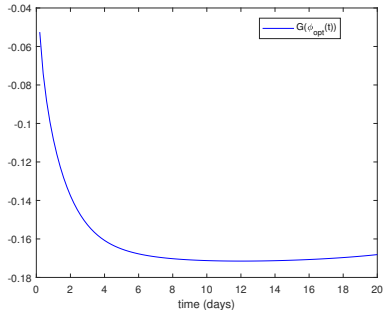
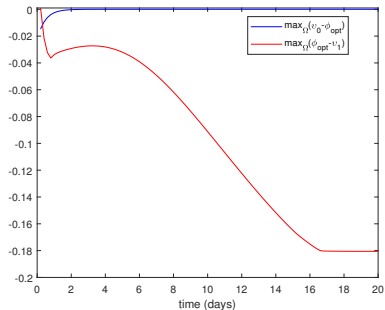


PART 2-1. Treatment of breast tumors

The cost parameters

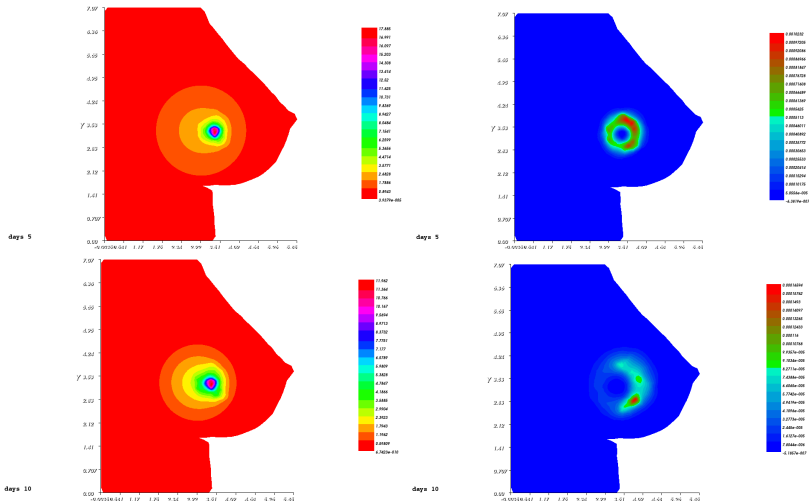
$$\lambda = 0.5 \cdot 10^{-2}; \beta_1 = 10^3; \beta_2 = 2 \cdot 10^4; \beta_3 = 10^6.$$

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf

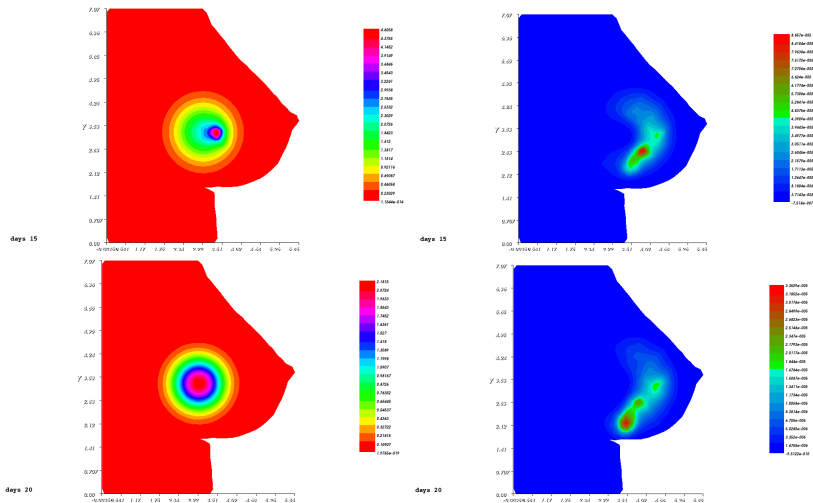


PART 2-1. Treatment of breast tumors

The optimal drug concentration ϕ_* and optimal tumor density u_* at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



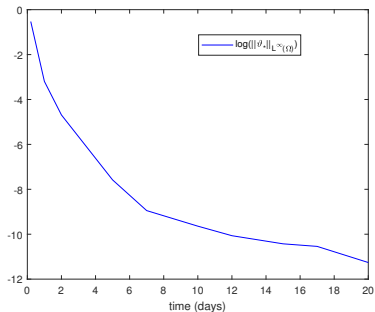
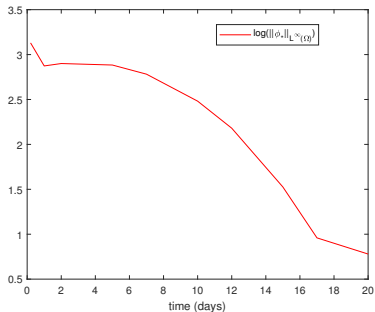
PART 2-1. Treatment of breast tumors



Time (t)	5^{th} days	10^{th} days	15^{th} days	20^{th} days
$\sup_{\Omega} \mathbf{u}_*$	0.00010	$6.5 \cdot 10^{-6}$	$4.96 \cdot 10^{-6}$	$3.35 \cdot 10^{-6}$
$\inf_{\Omega} \mathbf{u}_*$	$6.13 \cdot 10^{-8}$	$2.36 \cdot 10^{-8}$	$3.2 \cdot 10^{-8}$	$9.51 \cdot 10^{-11}$

PART 2-1. Treatment of breast tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale



PART 2-1. Treatment of breast tumors

(2) Case of two tumors

$$\mathbf{u}_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} + \eta_1 e^{-\frac{1}{\varepsilon}[(x-\tilde{x}_0)^2+(y-\tilde{y}_0)^2]}, \text{ in } \Omega, \quad (22)$$

the tumor cell density centered at (x_0, y_0) and $(\tilde{x}_0, \tilde{y}_0)$.

The discretization and mesh informations

Parameters	Value	Definitions
n_e	3310	Total mesh elements
n_p	1716	Total mesh points
$(x_0; y_0)$	(10/3; 2.12)	First Tumor center
$(\tilde{x}_0; \tilde{y}_0)$	(4.03; 2.3)	Second Tumor center
$(a; b)$	(3.68; 2.3)	
ε	0.02	

PART 2-1. Treatment of breast tumors

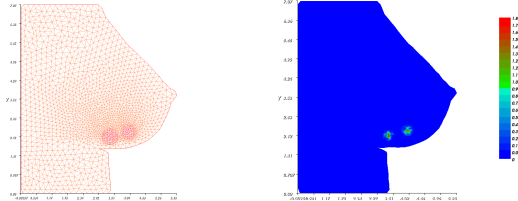
(2) Case of two tumors

$$\mathbf{u}_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} + \eta_1 e^{-\frac{1}{\varepsilon}[(x-\tilde{x}_0)^2+(y-\tilde{y}_0)^2]}, \text{ in } \Omega, \quad (22)$$

the tumor cell density centered at (x_0, y_0) and $(\tilde{x}_0, \tilde{y}_0)$.

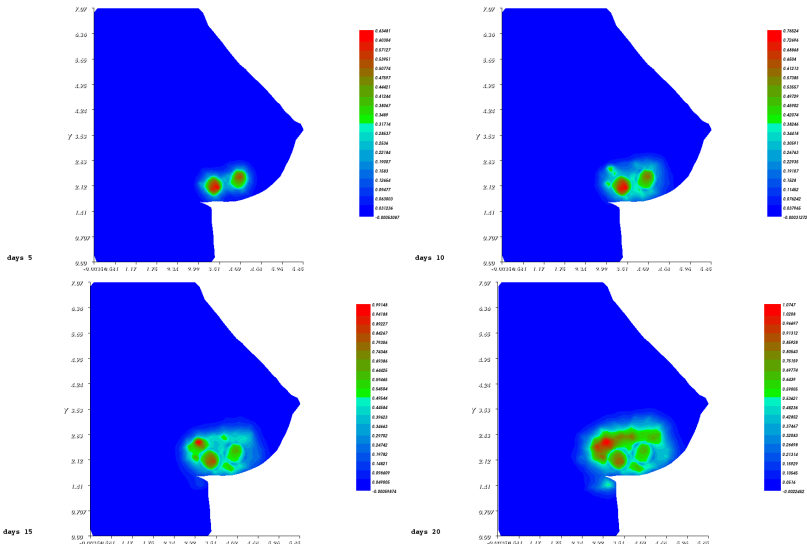
The discretization and mesh informations

Parameters	Value	Definitions
n_e	3310	Total mesh elements
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$(x_0; y_0)$	(10/3; 2.12)	First Tumor center
$(\tilde{x}_0; \tilde{y}_0)$	(4.03; 2.3)	Second Tumor center
$(a; b)$	(3.68; 2.3)	
ε	0.02	



PART 2-1. Treatment of breast tumors

The density of tumor cells $\mathbf{u}$ without treatment *i.e.*, $\phi = 0$ in (1) at 5^{th} , 10^{th} , 15^{th} , 20^{th} days

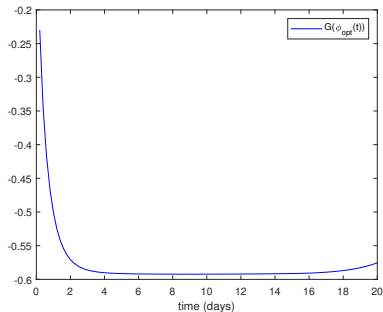
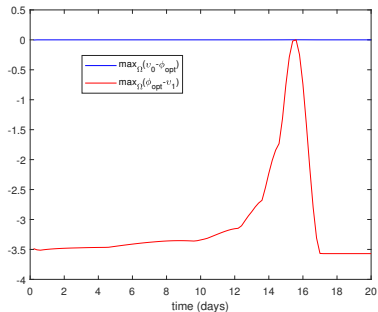


PART 2-1. Treatment of breast tumors

The cost parameters

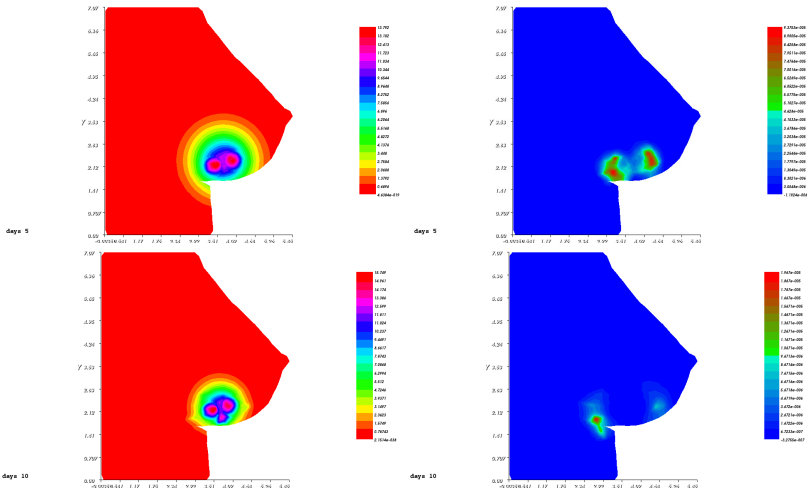
$$\lambda = 10^{-2}; \beta_1 = 2 \cdot 10^3; \beta_2 = 10^4; \beta_3 = 3 \cdot 10^6.$$

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf

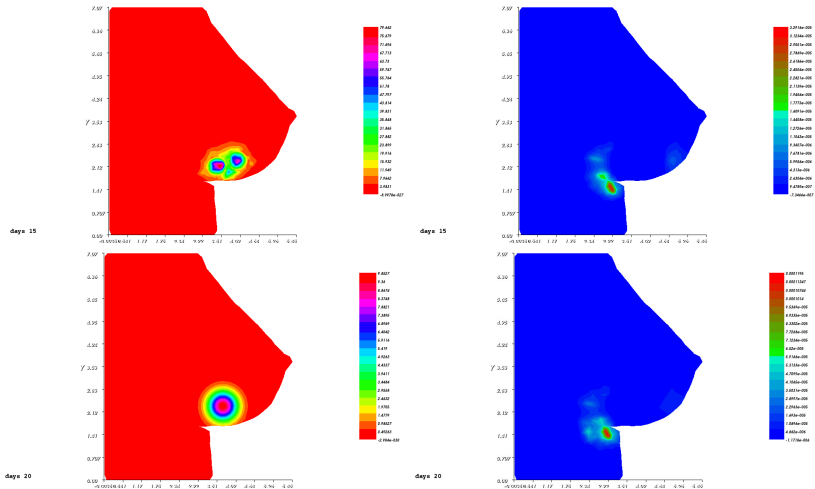


PART 2-1. Treatment of breast tumors

The optimal drug concentration ϕ_* and optimal tumor density u_* at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



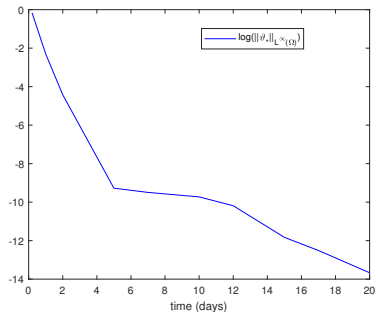
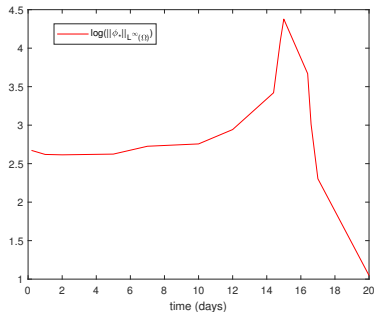
PART 2-1. Treatment of breast tumors



Time (t)	5 th days	10 th days	15 th days	20 th days
$\sup u_*$	0.00029	$5.22 \cdot 10^{-5}$	$3.34 \cdot 10^{-5}$	$2.77 \cdot 10^{-5}$
$\inf u_*$	$3.77 \cdot 10^{-7}$	$1.03 \cdot 10^{-7}$	$2.32 \cdot 10^{-7}$	$3.7 \cdot 10^{-7}$

PART 2-1. Treatment of breast tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale



PART 2-2. The influence of state constraint

The cost functional:

$$\mathcal{J}(\phi) = \frac{1}{2} \int_{\mathcal{Q}} \mathbf{u}^2 d\mathbf{x}dt + \frac{\lambda}{3} \int_{\mathcal{Q}} |\phi|^3 d\mathbf{x}dt, \quad (23)$$

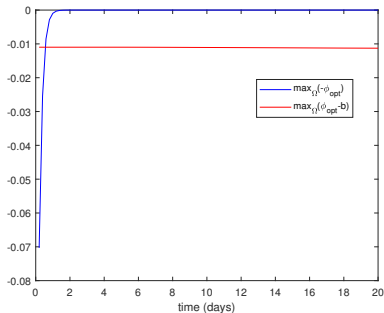
The control problem

$$\begin{aligned} & \inf_{\phi \in \mathcal{U}_{ad}} \mathcal{J}(\phi); \\ & s.t., \mathbf{u} = \mathcal{S}(\phi). \end{aligned} \quad (24)$$

PART 2-2. The influence of state constraint

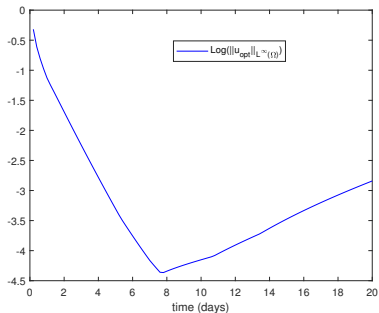
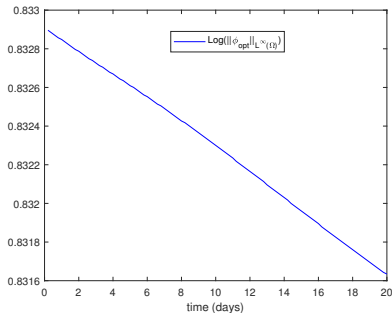
The cost parameters
 $\lambda = 10^{-3}$, $\beta_1 = 5 \cdot 10^4$, and $\beta_2 = 10^3$.

Constraints satisfaction of the optimal control ϕ_* in the following



PART 2-2. The influence of state constraint

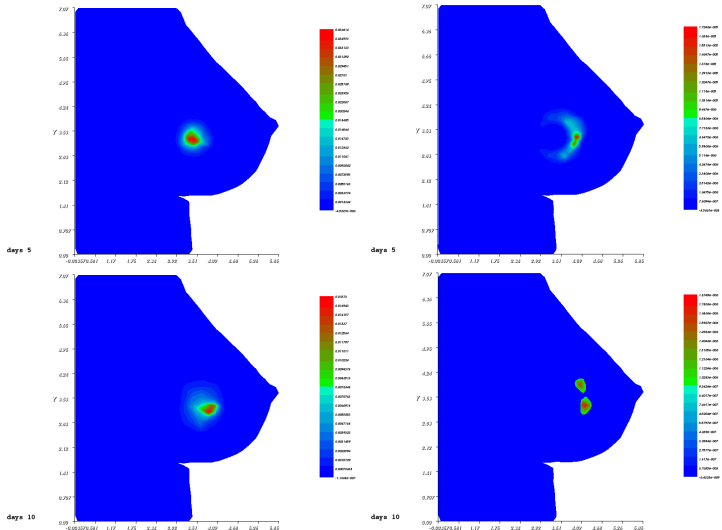
Evolution of the maximum value of φ_* on the left and $\mathbf{u}_*$ on the right in the logarithm scale.



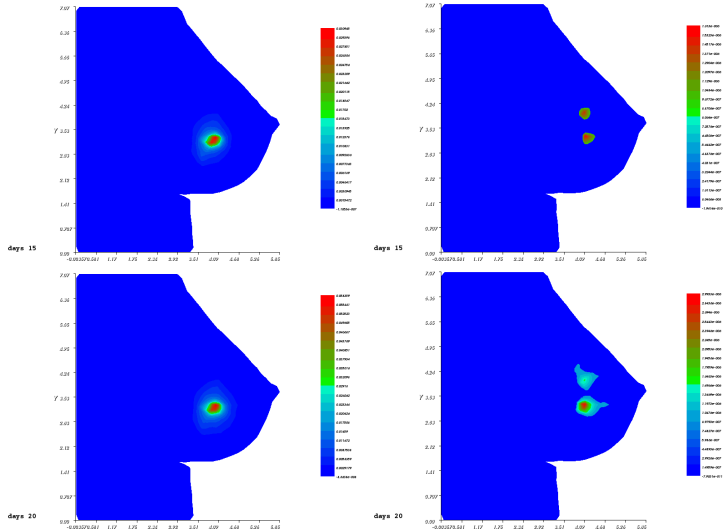
Time (t)	5th days	10th days	15th days	20th days
$\max_{\Omega} \mathbf{u}_*$	0.036	0.016	0.031	0.058
$\min_{\Omega} \mathbf{u}_*$	$4.52 \cdot 10^{-6}$	$1.13 \cdot 10^{-7}$	$1.11 \cdot 10^{-7}$	$8.86 \cdot 10^{-8}$

PART 2-2. The influence of state constraint

The optimal tumor density without state constraint on the left and the optimal tumor density with state constraint on the right.



PART 2-2. The influence of state constraint



PART 2-3. 3D Numerical simulations

$$u_0(\mathbf{x}) = ((x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2)e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]} \text{ in } \Omega,$$

The discretization and mesh informations

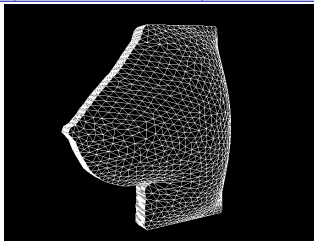
Parameters	Value	Definitions
$\mathbf{x}$	(x, y, z)	3D space position
Δx	1 mm	Space Discretizations (x)
Δy	1 mm	Space Discretizations (y)
Δz	1 mm	Space Discretizations (z)
n_e	52980	Total Mesh Elements
n_p	10384	Total Mesh Points
$(x_0; y_0; z_0)$	(2.34; 3.54; -1.75)	Tumor center
$(x_1; y_1; z_1)$	(2.3; 3.40; -1.70)	

PART 2-3. 3D Numerical simulations

$$u_0(\mathbf{x}) = ((x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2)e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]} \text{ in } \Omega,$$

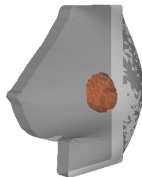
The discretization and mesh informations

Parameters	Value	Definitions
$\mathbf{x}$	(x, y, z)	3D space position
Δx	1 mm	Space Discretizations (x)
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Δz	1 mm	Space Discretizations (z)
n_e	52980	Total Mesh Elements
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$(x_0; y_0; z_0)$	(2.34; 3.54; -1.75)	Tumor center
$(x_1; y_1; z_1)$	(2.3; 3.40; -1.70)	



PART 2-3. 3D Numerical simulations

The density of tumor cells $\mathbf{u}$ without treatment *i.e.*, $\phi = 0$ in (1) at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



days 5

days 10

PART 2-3. 3D Numerical simulations



days 15



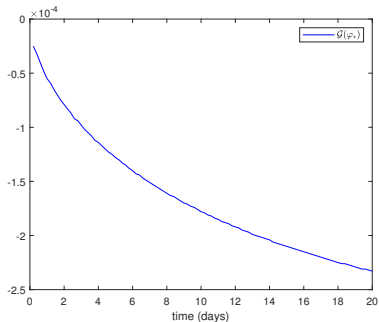
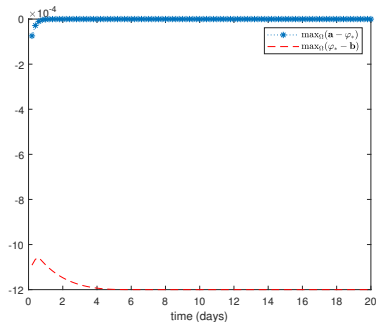
days 20

PART 2-3. 3D Numerical simulations

The penalized cost parameters

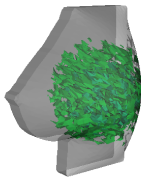
$$\lambda = 10^{-2}; \beta_1 = 10^5; \beta_2 = 5 \cdot 10^4; \beta_3 = 10^4.$$

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf

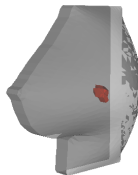


PART 2-3. 3D Numerical simulations

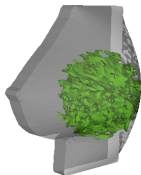
The optimal drug concentration ϕ_* and optimal tumor density u_* at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



days 5



days 5

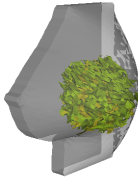


days 10

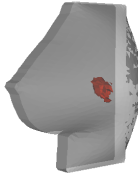


days 10

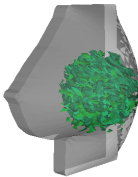
PART 2-3. 3D Numerical simulations



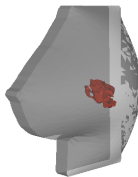
days 15



days 15



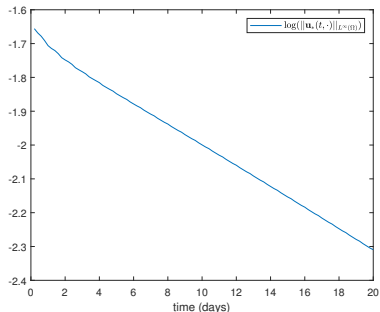
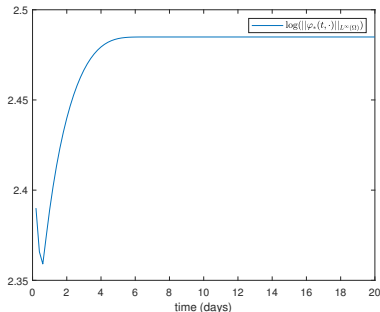
days 20



days 20

PART 2-3. 3D Numerical simulations

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale



Time (t)	5th days	10th days	15th days	20th days
$\max_{\Omega} \mathbf{u}_*$	0.01587	0.01362	0.01160	0.009130
$\min_{\Omega} \mathbf{u}_*$	0.003261	0.004813	0.005233	0.005079

PART 2-4. Stochastic mathematical model

$$\left\{ \begin{array}{l} d\mathbf{u} = (\operatorname{div}(D(t, \mathbf{x}; \omega) \nabla \mathbf{u}) + K_2(t, \mathbf{x}; \omega) \mathbf{u} - K_1(t, \mathbf{x}; \omega) \mathbf{u}^3) dt \\ \quad + \sigma_0 K_3(t, \mathbf{x}; \omega) \mathbf{u} dW_1(t) - \alpha_0 \phi(t, \mathbf{x}; \omega) \mathbf{u} dW_2(t) \\ \quad \text{in } (0, T) \times \mathcal{O} \times \Omega, \\ D(t, \mathbf{x}; \omega) \nabla \mathbf{u} \cdot \mathbf{n} = 0 \text{ on } \Sigma \times \Omega, \\ \mathbf{u}(0, \mathbf{x}; \omega) = u_0(\mathbf{x}, \omega) \text{ in } \mathcal{O} \times \Omega, \end{array} \right. \quad (25)$$

under the pointwise constraint

$$0 \leq \varphi(t, \mathbf{x}; \omega) \leq \mathbf{b}(\mathbf{x}, \omega), \text{ a.e. } (t, \mathbf{x}; \omega) \in \mathcal{Q} \times \Omega. \quad (26)$$

PART 2-4. Stochastic control problem

The cost functional:

$$\mathcal{J}(\varphi) := \mathbb{E}_{\Omega} \left(\frac{1}{2} \int_{\mathcal{Q}} |\mathbf{u}(t, \mathbf{x}; \omega)|^2 \, d\mathbf{x} \, dt + \frac{\lambda}{3} \int_{\mathcal{Q}} |\phi(t, \mathbf{x}; \omega)|^3 \, d\mathbf{x} \, dt \right). \quad (27)$$

PART 2-4. Stochastic control problem

The cost functional:

$$\mathcal{J}(\varphi) := \mathbb{E}_{\Omega} \left(\frac{1}{2} \int_{\mathcal{Q}} |\mathbf{u}(t, \mathbf{x}; \omega)|^2 d\mathbf{x} dt + \frac{\lambda}{3} \int_{\mathcal{Q}} |\phi(t, \mathbf{x}; \omega)|^3 d\mathbf{x} dt \right). \quad (27)$$

The control problem **(CP)**:

$$\begin{aligned} & \inf_{\phi \in \mathcal{U}_{ad}} \mathcal{J}(\phi); \\ & \text{s.t. } \mathcal{G}(\mathbf{u}) = \|\mathbf{u}(t, \cdot)\|_{L^2(\mathcal{O})}^2 - \zeta \leq 0, \text{ a.s.}, \end{aligned} \quad (28)$$

where

$$\mathcal{U}_{ad} = \{\phi \in L^3(\Omega; L^3(\mathcal{Q})), 0 \leq \varphi(t, \mathbf{x}; \omega) \leq \mathbf{b}(\mathbf{x}, \omega), \text{ a.e. } t \in [0, T], \text{ a.s. } \omega \in \Omega\}.$$

PART 2-4. Preliminary results

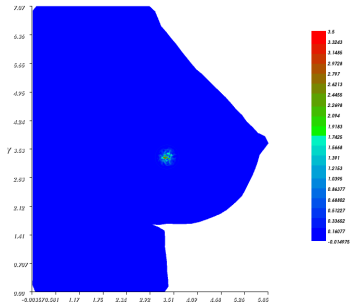
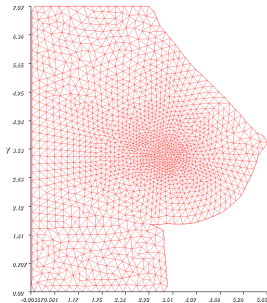
Let $\mathbf{U}$ be a uniform random distribution on $[0, 1]$, for $(t, \mathbf{x}) \in (0, T) \times \bar{\mathcal{O}}$:

$$\mathbf{D} = \mathbf{D}_0 e^{-\lambda_0 \sigma} \cdot \mathbf{U}; \quad \sigma = \varrho_0 e^{-\delta_0 |\mathbf{x}|^2 t}; \quad \mathbf{K}_1 = \frac{k_1 - k_2 e^{-\gamma_0 \sigma}}{e^{-\gamma_0 \sigma} (1 - e^{-2\gamma_0 \sigma})} \cdot \mathbf{U}; \quad \mathbf{K}_2 = \theta^2 \mathbf{K}_1;$$

$$\mathbf{K}_3 = k_3 \cdot \mathbf{U}; \quad \mathbf{D}_0 = d_0 \eta_1; \quad \mathbf{b} = \gamma_0 e^{-\gamma_1 [(x-x_1)^2 + (y-y_1)^2]} \cdot \mathbf{U}; \quad \zeta = \|u_0\|_{L^2(\mathcal{O} \times \Omega)}^2$$

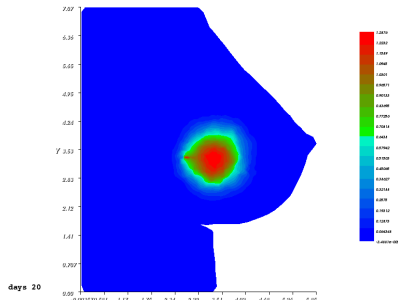
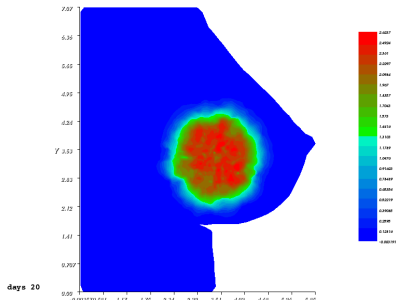
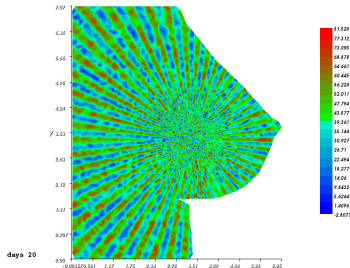
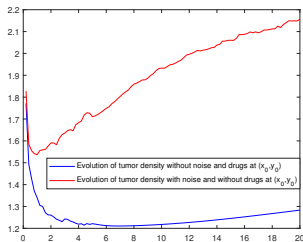
$$\xi_1 := \frac{dW_1}{dt} = (1 + \cos(\eta_2)) \frac{e^{-(\frac{t}{\tau})^2}}{\tau \sqrt{2\pi}}; \quad \xi_2 = \sigma_0 \xi_1.$$

We consider the initial tumor density $u_0(\mathbf{x}, \omega) = 2\eta_1 e^{-\frac{1}{\varepsilon} [(x-x_0)^2 + (y-y_0)^2]} \cdot \mathbf{U}(\omega)$



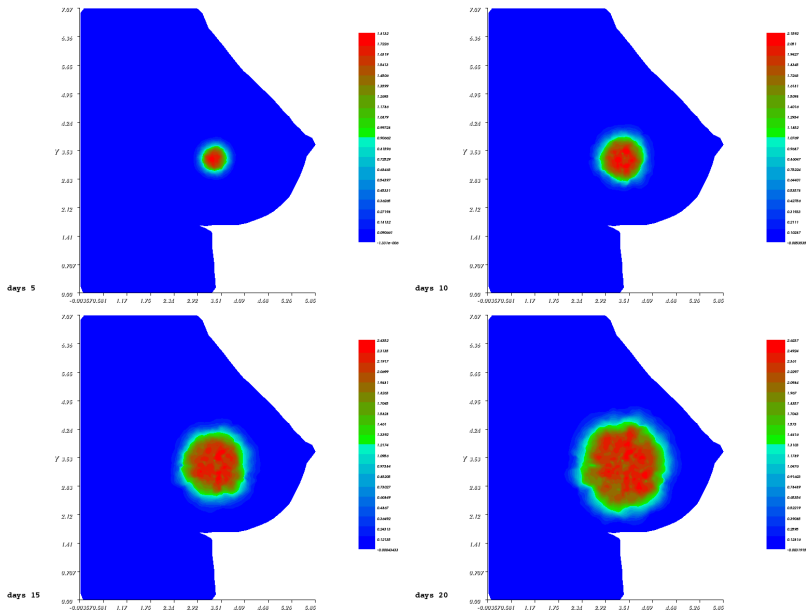
PART 2-4. Preliminary results

The influence of noise on tumor density evolution without treatment on the left and the noise ξ_1 at 20th day on the right.



PART 2-4. Preliminary results

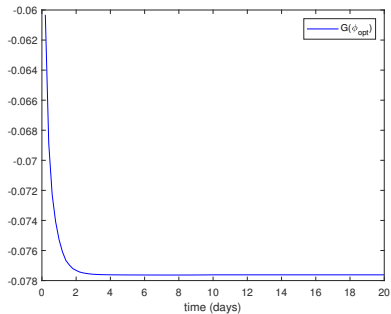
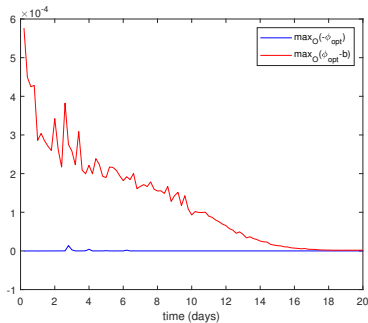
The density of tumor cells $\mathbf{u}$ without treatment *i.e.*, $\phi = 0$ in (25) at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



PART 2-4. Preliminary results

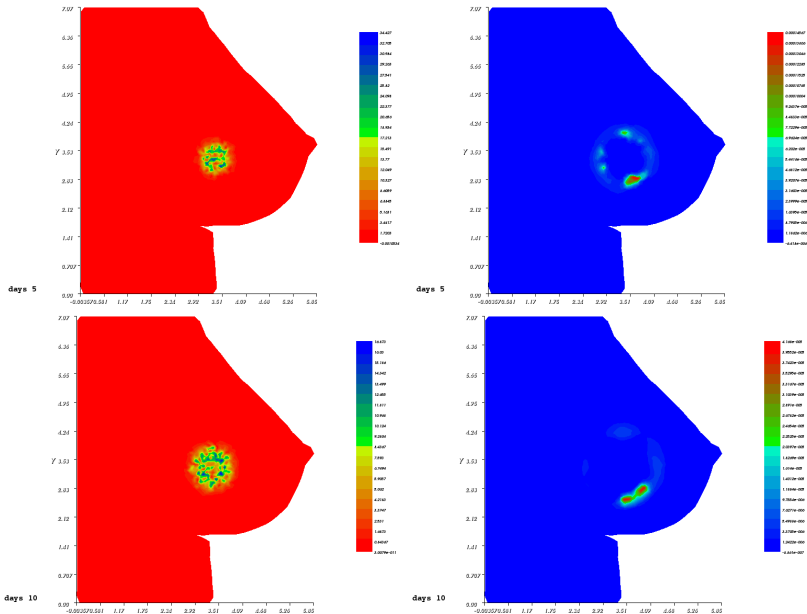
The penalised cost parameters
 $\lambda = 10^{-2}$, $\beta_1 = 10^4$, $\beta_2 = 3 \cdot 10^5$, $\beta_3 = 10^2$

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following

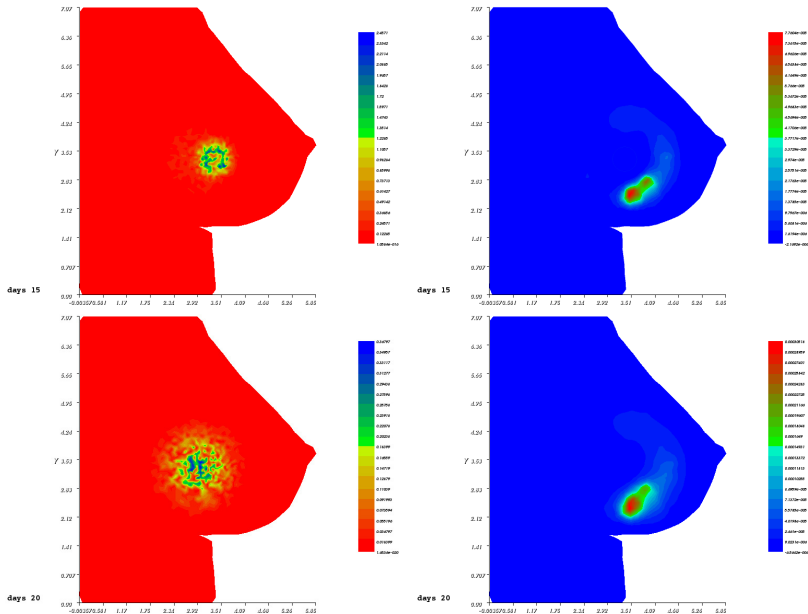


PART 2-4. Preliminary results

The optimal drug concentration ϕ_* and optimal tumor density u_* at 5^{th} , 10^{th} , 15^{th} , 20^{th} days

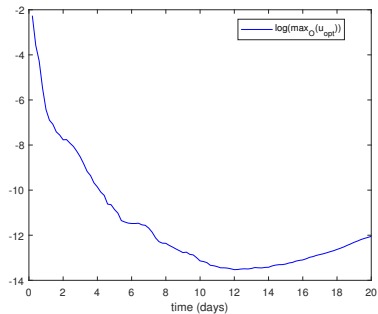
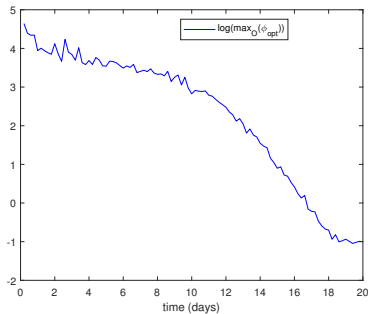


PART 2-4. Preliminary results



PART 2-4. Preliminary results

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale



PART 2-5. Treatment of lung tumors

$$D = D_0 e^{-\gamma_0 \sigma}; \quad \sigma = \sigma_0 e^{-\delta_0 |\mathbf{x}|^2 t}; \quad K_1 = k_1 e^{-\delta_0 |\mathbf{x}|^2 t}; \quad K_2 = \theta^2 K_1; \quad \mathbf{a}(\mathbf{x}) = 0;$$

$$\mathbf{b}(\mathbf{x}) = b_0 e^{-b_1 [(x-x_1)^2 + (y-y_1)^2]}; \quad K_3 = k_3 K_1 \Gamma; \quad D_0 = d_0 \eta_1; \quad \xi = \|u_0\|_{L^2(\Omega)}^2.$$

The discretization and mesh informations are the following

$$\Delta t = 0.2 \text{ days (the time step)}; \quad \mathbf{x} = (x, y); \Delta x = \Delta y = 1 \text{ mm};$$

$$n_e = 1492 \text{ (number mesh elements)}; \quad n_p = 821 \text{ (number mesh points)};$$

$$(x_0, y_0) = (1.34, 3.8) \text{ (the tumor center)}; \quad (x_1, y_1) = (1.3, 3.5).$$

Parameters	d_0	δ_0	k_1	α_0	σ_0
Values	$0.2 \text{ mm}^2/d$	$5.55 \cdot 10^{-5}/\text{mm}^2 \cdot d$	$0.44/d$	$1/\mu\text{M} \cdot d$	25 mmHg

γ_0	k_3	T	ς_1	ς_2	ς_3	ς_4	θ	λ	β_1	β_2	β_3
$1/\text{mmHg}$	3 mm	20 d	0.7	50	2	10^{-4}	2	10^{-2}	10^3	$3 \cdot 10^4$	$5 \cdot 10^6$

b_0	b_1
$100 \mu\text{M}$	11.11 mm^2

PART 2-5. Treatment of lung tumors

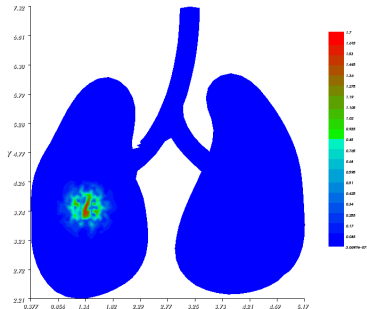
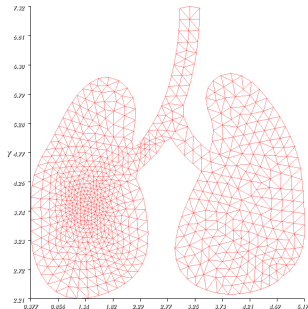
(1) Case of tumor in the left lung

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon} [(x-x_0)^2 + (y-y_0)^2]} \text{ in } \Omega. \quad (29)$$

PART 2-5. Treatment of lung tumors

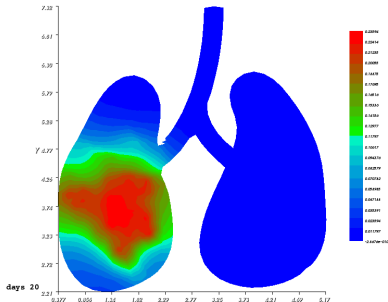
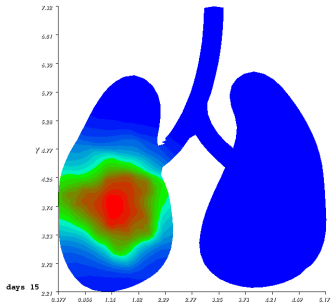
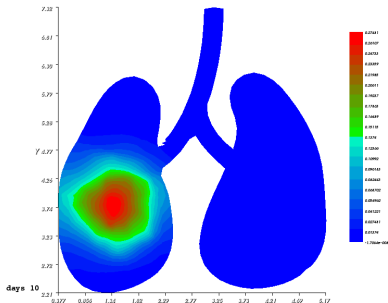
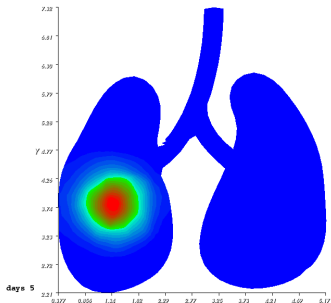
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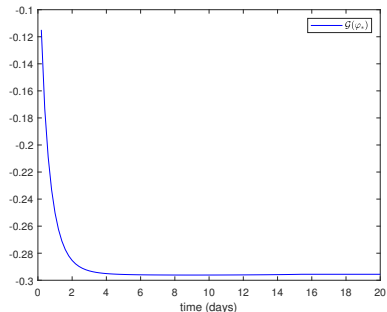
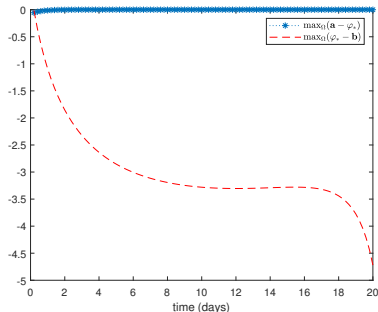
PART 2-5. Treatment of lung tumors

The density of tumor cells $\mathbf{u}$ without treatment *i.e.*, $\phi = 0$ in (1) at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



PART 2-5. Treatment of lung tumors

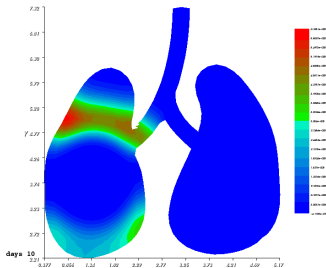
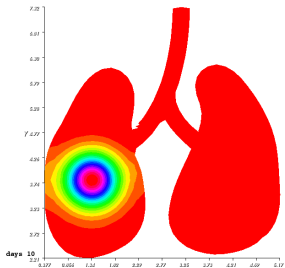
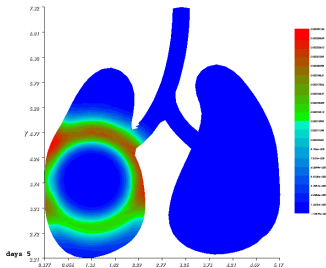
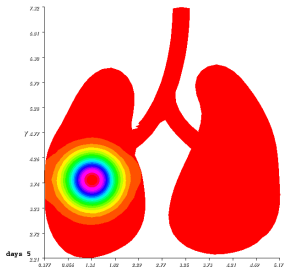
Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf



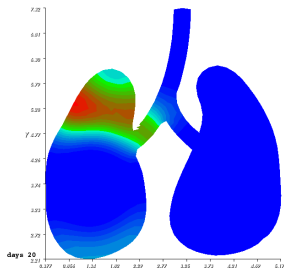
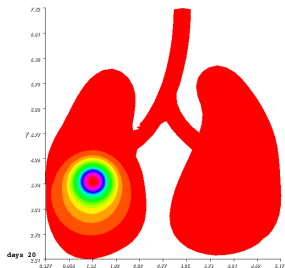
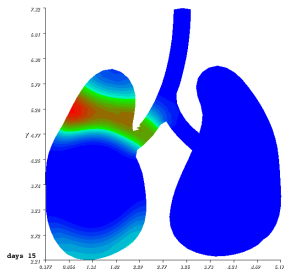
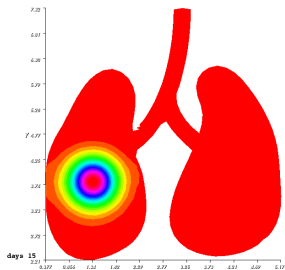
Time (t)	5th day	10th day	15th day	20th day
$\max_{\Omega} \mathbf{u}_*$	0.00045	$6.10 \cdot 10^{-5}$	$3.54 \cdot 10^{-5}$	$2.72 \cdot 10^{-5}$
$\min_{\Omega} \mathbf{u}_*$	$1.55 \cdot 10^{-7}$	$4.17 \cdot 10^{-10}$	$8.78 \cdot 10^{-11}$	$1.14 \cdot 10^{-12}$

PART 2-5. Treatment of lung tumors

The optimal drug concentration ϕ_* and optimal tumor density u_* at 5^{th} , 10^{th} , 15^{th} , 20^{th} days

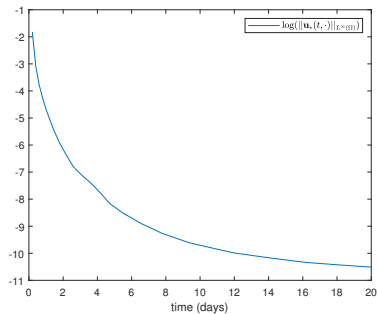
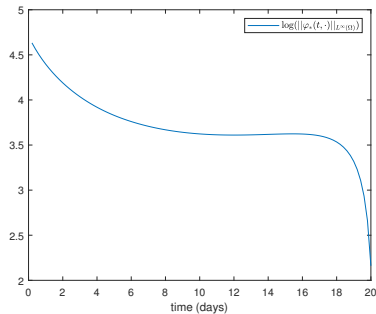


PART 2-5. Treatment of lung tumors



PART 2-5. Treatment of lung tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale



PART 2-5. Treatment of lung tumors

(2) Case of tumor in the right lung

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2 + (y-y_0)^2]} \text{ in } \Omega. \quad (30)$$

Mesh data and the parameters

(x_0, y_0)	(x_1, y_1)	ε	n_e	n_p	λ	β_1	β_2	β_3
(3.8, 4.5)	(1.3, 3.5)	0.1	1444	797	$5 \cdot 10^{-2}$	$2 \cdot 10^4$	10^5	10^4

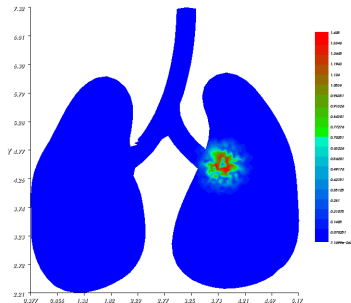
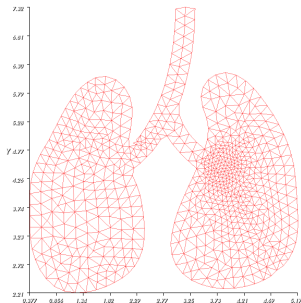
PART 2-5. Treatment of lung tumors

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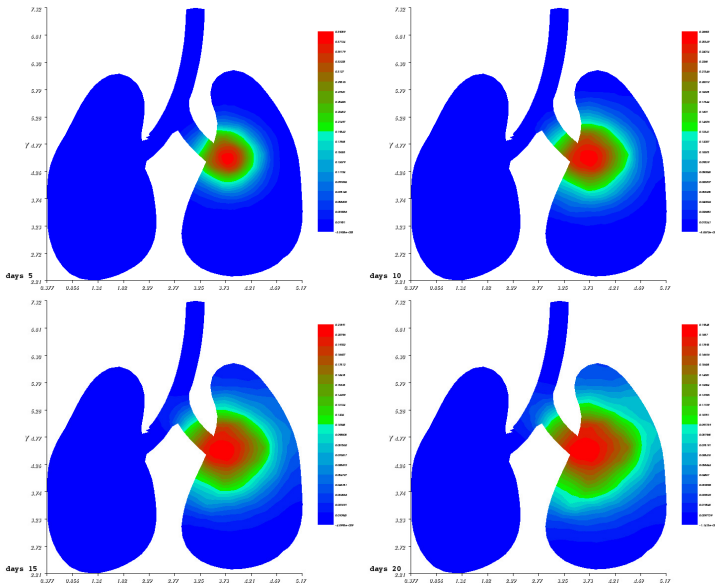
Mesh data and the parameters

(x_0, y_0)	(x_1, y_1)	ε	n_e	n_p	λ	β_1	β_2	β_3
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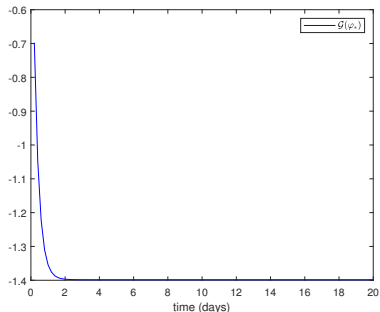
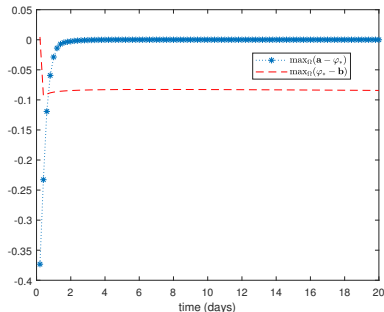
PART 2-5. Treatment of lung tumors

The density of tumor cells $\mathbf{u}$ without treatment *i.e.*, $\phi = 0$ in (1) at 5^{th} , 10^{th} , 15^{th} , 20^{th} days



PART 2-5. Treatment of lung tumors

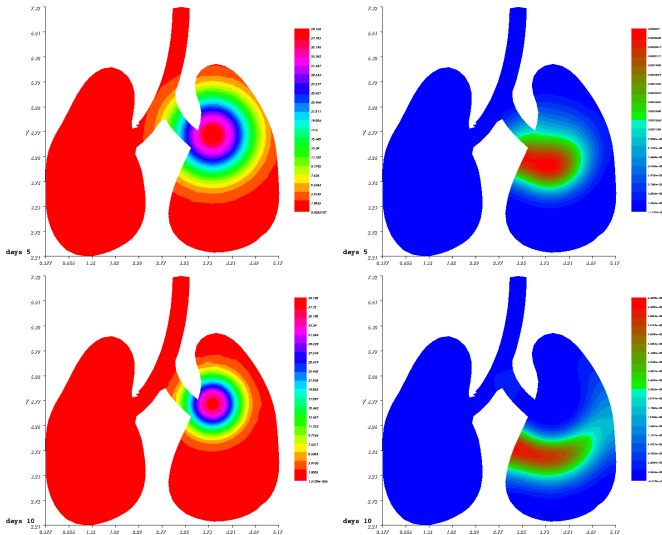
Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf



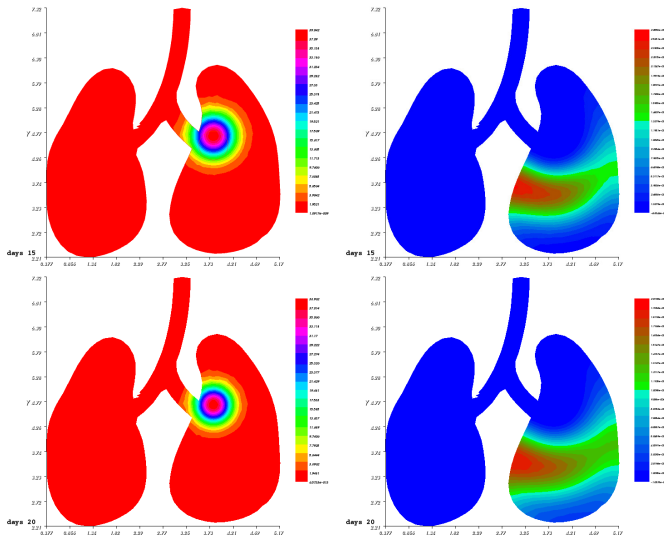
Time (t)	5th day	10th day	15th day	20th day
$\max_{\Omega} \mathbf{u}_*$	0.00067	$4.49 \cdot 10^{-5}$	$2.65 \cdot 10^{-5}$	$2.09 \cdot 10^{-5}$
$\min_{\Omega} \mathbf{u}_*$	$1.17 \cdot 10^{-7}$	$5.37 \cdot 10^{-11}$	$6.83 \cdot 10^{-12}$	$1.037 \cdot 10^{-12}$

PART 2-5. Treatment of lung tumors

The optimal drug concentration ϕ_* and optimal tumor density u_* at 5^{th} , 10^{th} , 15^{th} , 20^{th} days

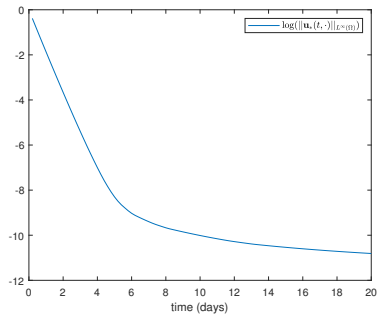
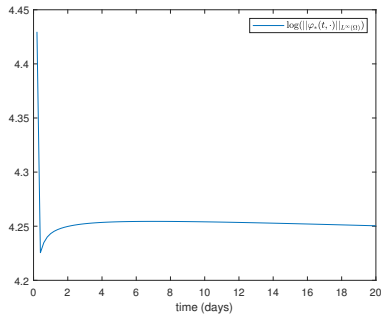


PART 2-5. Treatment of lung tumors



PART 2-5. Treatment of lung tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale



Outline of the talk

- 1 Mathematical modeling
- 2 Optimal control problem
- 3 Applications to realistic problems
 - Treatment of breast tumors
 - The influence of state constraint in treatment
 - Treatment of a tumor in a 3D breast domain
 - Treatment of breast tumors in the presence of noise data
 - Treatment of lung tumors
- 4 Conclusion and future work

Conclusion and future work

What is done ?

- New methodology for the traitement of breast and lung tumors.
- The simulation results prove the importance of the constraints and are promising.

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Current and Future Work

- Mathematical analysis of the stochastic control problem.
- Integrating random or noisy data into lung tumor treatment.
- System coupled with normal cell evolution.

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- Mathematical analysis of the stochastic control problem.
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Open problem!!!



Thank you for your attention!!!